Reconstructing the MgO thermal EOS

Equations and Debye-temperature prescription in Speziale (2001)

Purpose. Reconstruct the thermal EOS step by step, starting from the 300 K reference curve. Sergio Speziale has confirmed the Debye-temperature integration used for the final variable-q model (see Section 7). All equations needed for our calculation are written out here. Labels S1–S4, S10 and S11 correspond directly to the same equation numbers in Speziale et al. (2001), expressed in the notation below. Labels N1–N11 identify equations or derivations added in this note.

1. Reference state and normalization

For γ_0, we use the value explicitly fixed in Section 5.1 of the paper: $\gamma_0 = 1.524$. The Figure 6 caption reports $\gamma_0 = 1.524 \pm 0.025$. All calculations below use this printed central value; γ_0 is not refitted.

Quantity	Adopted value	Definition
V_0	74.71 Å^3	Conventional-cell volume at T_r, zero pressure
K_0	160.2 GPa	Isothermal bulk modulus at the reference state
K_0'	3.99	$(\partial K_T/\partial P)_{T_r}$ at zero pressure
γ_0	1.524	Published fixed value (Section 5.1)
$q_0,\ q_1$	1.65, 11.8	Constants in the variable-q model
θ_0	773 K	Reference Debye temperature
T_r	300 K	Principal isotherm and energy subtraction
$Z,\ n$	4, 2	Formula units per cell; atoms per MgO formula unit

V_c denotes the numerical conventional-cell volume in Å^3; V_m is the molar volume in m^3 per mole of MgO. Define

$$x = \frac{V_c}{V_0}, \qquad V_m = \frac{N_A}{Z}\, V_c\, 10^{-30}, \qquad N_A = 6.02214076 \times 10^{23}\ \text{mol}^{-1}. \tag{N1}$$

This unit conversion is explicit here; it is not a numbered equation in the paper. The gas constant is $R = 8.31446261815324\ \text{J mol}^{-1}\,\text{K}^{-1}$.

2. Complete pressure calculation

Sections 2.1 and 4.2 specify a third-order Birch–Murnaghan reference isotherm. We write its standard expression explicitly:

$$P_{300}(V_c) = \frac{3K_0}{2}\left(x^{-7/3} - x^{-5/3}\right)\left[1 + \frac{3}{4}(K_0' - 4)\left(x^{-2/3} - 1\right)\right]. \tag{N2}$$

With K_0 in GPa, this expression returns GPa. For the Debye energy, the paper's Equation 3 becomes, with $y = \theta(V_c)/T$,

$$E_D(V_c, T) = \frac{9nRT}{y^3}\int_0^y \frac{z^3}{e^z - 1}\, dz. \tag{S3}$$

E_D is in J per mole of MgO; $n = 2$. The same $\theta(V_c)$ is used at both temperatures in the fixed-volume subtraction. The paper's Equations 2 and 1 then give the thermal increment and total pressure, respectively:

$$\Delta P_{\text{th}}(V_c, T) = 10^{-9}\frac{\gamma(V_c)}{V_m}\left[E_D(V_c, T) - E_D(V_c, T_r)\right], \tag{S2}$$

$$P(V_c, T) = P_{300}(V_c) + \Delta P_{\text{th}}(V_c, T). \tag{S1}$$

The factor 10^{-9} converts Pa to GPa. Thus $P(V_0, 300\ \text{K}) = 0$. The electronic contribution is omitted, as in the MgO model under discussion; any temperature-independent vibrational term cancels in the energy difference.

3. Volume dependence of the Grüneisen parameter

Section 2.1 defines q through the logarithmic volume derivative of γ. We write that unnumbered definition as

$$q(V_c) = \left(\frac{\partial \ln \gamma}{\partial \ln V_c}\right)_T . \tag{N3}$$

The paper's final variable-q model (its Equations 10–11) is

$$q(V_c) = q_0 x^{q_1}, \tag{S10}$$

$$\gamma(V_c) = \gamma_0 \exp\left[\frac{q_0}{q_1}\left(x^{q_1} - 1\right)\right]. \tag{S11}$$

Equation (S11) follows by integrating Equation (N3) with $\gamma(V_0) = \gamma_0$. It also gives a finite value at infinite compression:

$$\lim_{x\to 0} \gamma(x) = \gamma_0 e^{-q_0/q_1} = 1.325127 \quad \text{for the adopted coefficients.} \tag{N4}$$

4. Debye-temperature calculation and the question

We reconstruct the Debye temperature from the Debye–Grüneisen relation below. This is the thermodynamic relation used for our derivation, rather than an explicit numbered equation in the paper:

$$\gamma(V_c) = -\frac{d \ln \theta(V_c)}{d \ln V_c}, \qquad \theta(V_0) = \theta_0. \tag{N5}$$

For **constant** $q \neq 0$, integration gives

$$\gamma(V_c) = \gamma_0 x^q, \tag{N6}$$

$$\theta(V_c) = \theta_0 \exp\left[\frac{\gamma_0 - \gamma(V_c)}{q}\right]. \tag{S4}$$

The second expression is the paper's Equation 4; the first is the constant-q gamma law discussed in Section 5.4. For $q = 0$, the corresponding limit is $\gamma(V_c) = \gamma_0$ and $\theta(V_c) = \theta_0 x^{-\gamma_0}$.

For the **variable-q** model, our implementation instead integrates Equation (N5) using Equation (S11):

$$\boxed{\theta(V_c) = \theta_0 \exp\left[-\int_1^x \gamma_0 \exp\left(\frac{q_0}{q_1}(u^{q_1} - 1)\right)\frac{du}{u}\right].} \tag{N7}$$

An independent closed form, with $a = q_0/q_1$, is

$$\ln \frac{\theta(V_c)}{\theta_0} = -\frac{\gamma_0 e^{-a}}{q_1}\left[\mathrm{Ei}(a x^{q_1}) - \mathrm{Ei}(a)\right]. \tag{N8}$$

Here Ei is the exponential integral, defined on positive arguments by $\mathrm{Ei}(z) = \mathrm{PV}\int_{-\infty}^{z} e^t/t\,dt$, where PV denotes the Cauchy principal value. Its derivative is $d\,\mathrm{Ei}(z)/dz = e^z/z$. Equation (N8) is used for the adopted positive q_0 and q_1.

Diagnostic illustration, not a known author prescription. Simply inserting instantaneous $q(V_c)$ into Equation (S4) would give the hypothetical expression

$$\theta_{\mathrm{local}}(V_c) = \theta_0 \exp\left[\frac{\gamma_0 - \gamma(V_c)}{q(V_c)}\right]. \tag{N9}$$

For Equations (S10)–(S11), differentiating this candidate gives

$$-\frac{d \ln \theta_{\mathrm{local}}}{d \ln V_c} = \gamma(V_c) + \frac{q_1}{q(V_c)}\left[\gamma_0 - \gamma(V_c)\right]. \tag{N10}$$

The extra term arises because $q(V_c)$ also changes with volume. This *hypothetical expression*, Equation (N9), therefore does not satisfy Equation (N5) in general. Our integrated Equation (N7) does satisfy that relation. Speziale's reply now confirms that this integration was used in his original calculation.

5. Reference calculations

These **Peritheos calculations** use the equations on pages 1–2 and the published central parameter values. They are model outputs, not measured or transcribed observations. The digits facilitate numerical comparison and do not indicate physical parameter accuracy.

V_c/V_0	V_c (Å^3)	T (K)	γ	θ (K)
1.00	74.71000	300	1.524000	773.000
1.00	74.71000	1100	1.524000	773.000
0.90	67.23900	1100	1.379666	899.275
0.80	59.76800	2000	1.338508	1054.804
0.64	47.81440	3663	1.326084	1419.208

V_c/V_0	T (K)	P_{300} (GPa)	$\Delta P_{\rm th}$ (GPa)	P (GPa)
1.00	300	0.000000	0.000000	0.000000
1.00	1100	0.000000	4.964240	4.964240
0.90	1100	20.830878	4.855028	25.685906
0.80	2000	55.839883	11.610349	67.450232
0.64	3663	174.738015	28.659685	203.397700

6. Checks and their limits

Equation and normalization checks. We verify the defining derivatives in Equations (N3) and (N5). At $T = T_r$, the thermal pressure is zero at every volume, and $P(V_0, T_r) = 0$. Evaluating the Debye energy per mole of MgO ($n = 2$, $Z = 4$) or per mole of conventional cells ($n = 8$, $Z = 1$) gives the same pressure. The high-temperature limit is

$$\lim_{T/\theta \to \infty} \frac{E_D(V_c, T)}{T} = 3nR = 6R \quad \text{per mole of MgO.} \tag{N11}$$

Independent numerical evaluation. With all coefficients fixed to the published central values, the 64-point Gauss–Legendre calculation, independently coded adaptive integration and the closed form in Equation (N8) agree to better than 10^{-9} GPa on 35 states spanning $x = 0.60$–1.02 and $T = 300$–3663 K. This checks numerical evaluation of the stated equations; it is not a claim of physical accuracy at that pressure precision.

Comparison with the paper's calculated curves. Figure 6 tests the constant-q branch ($q = 1.65$), using Equation (S4). Eleven selected checkpoints on the 300, 1100 and 3000 K curves differ by at most approximately 0.11 GPa, or 0.0004 in V_c/V_0. The Figure 10 inset tests the final variable-q branch: four selected checkpoints on its 1100 K curve differ by at most approximately 0.30 GPa, or 0.0012 in V_c/V_0. These are comparisons with digitized *calculated curves*, not refits to measurements. Scan resolution, axis placement, dotted-line phase and stroke width limit this comparison. The integration prescription is now author-confirmed; these graphical differences do not recover the original numerical settings.

Remaining scope. The complete original thermal fitting procedure and shock-data reduction have not been reproduced. The integration prescription is confirmed, but the original calculator, trapezoidal grid and numerical settings remain unavailable.

7. Author confirmation recorded on 9 October 2026

In the reply supplied by Clemens, Sergio Speziale states: *"I can confirm that I numerically integrated your equation N5 by using S11, using the equation numbering of your notes."* He adds: *"I generally used a simple trapezoidal integration approach."* He cannot currently retrieve the original calculator. This confirms our continuous integration prescription, while leaving exact historical numerical replay and the complete author fit unestablished.

Primary reference. Speziale, Zha, Duffy, Hemley and Mao (2001), *Quasi-hydrostatic compression of magnesium oxide to 52 GPa: Implications for the pressure-volume-temperature equation of state*, J. Geophys. Res. 106, 515-528. doi:10.1029/2000JB900318. Source model: Section 2.1, Equations 1-4; Sections 4.2 and 5.1-5.4, Equations 10-11; Figures 6, 9 and 10.